MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

(D) 1/ TYPE 250TH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode

See figure 1

Weight: 10 ounces nominal (283.5 grams)

Mounting position: Vertical

ABSOLUTE RATINGS: C Telegraphy

Parameter:	F	Ef	Eb	Ec	Ib	Ic	Pp
Unit:	MHz	V	kVdc	Vdc	mAdc	mAdc	W
Maximum:	40	5.25	3.0	-210	330	75.0	250
Minimum:	---	4.75	---	---	---	---	---
TEST CONDITIONS:	---	5.0 Vac	2.5	Adjust	100	---	---

GENERAL:

Qualification - Required

Holding period (MIL-STD-1311) - See note 1

1/ See note 2

(D) denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1301	Filament current		0.65	II	If	9.7	11.2	Aac
1266	Total grid current		0.65	II	Ic	---	-15	µAdc
1261	Electrode voltage (grid)		0.65	II	Ec	-27.0	-39.0	Vdc
1266	Primary grid emission	Ef = 5.5 Vac; Ic = 175 mAdc; t = 15	0.65	II	Isg	---	-500	µAdc
1231	Peak emission	eb = ec = 2,500 v	0.65	II	is	5.0	---	a
	Quality conformance inspection, part 2							
1316	Amplification factor		---	---	Mu	31.0	41.0	---
1236	Power oscillation	Eb = 3,000 Vdc; Ib = 300 mAdc; F = 8 MHz	---	---	Po	540	---	W
1331	Direct-interelectrode capacitance		---	---	Cgp Cin Cout	2.2 3.7 ---	3.0 5.1 0.6	pF pF pF
	Quality conformance inspection, part 3							
---	Life test	Group C; power oscillation; t = 500 hours	---	---	---	---	---	---
---	Life-test end points:							
1231	Peak emission		---	---	is	3.5	---	a
1266	Primary grid emission		---	---	Isg	---	-500	µAdc
1031	Low-frequency vibration	No voltages (see note 3)	---	---	---	---	---	---
1036	Bump	Angle = 20° (see note 3)	---	---	---	---	---	---
---	Low-frequency vibration and bump end points:							
1266	Total grid current		---	---	Ic	---	-15	µAdc
1261	Electrode voltage (grid)		---	---	Ec	-27.0	-39.0	Vdc

NOTES:

1. In addition to the 24-hour holding period, 10 percent of the tubes in any production lot shall be held for a period of 72 hours. If more than 2 percent of the tubes fail to meet the total grid current test, the tubes represented by this sample shall be subject to recall and all tubes shall be held for a period of 72 hours.
2. Tube type 250TL has been deleted from this tube specification sheet. Use EIA Type 250TL.
(D) 3. This test shall be performed yearly, with sampling as follows:

n1 = 4 c1 = 0 where c2 represents the total allowable failures for
n2 = 4 c2 = 1; the first and second samples combined.

In case of failure, the test shall become a quality conformance inspection (QCI), part 2 test for three successful successive submissions, at which time the test may revert to the yearly basis.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

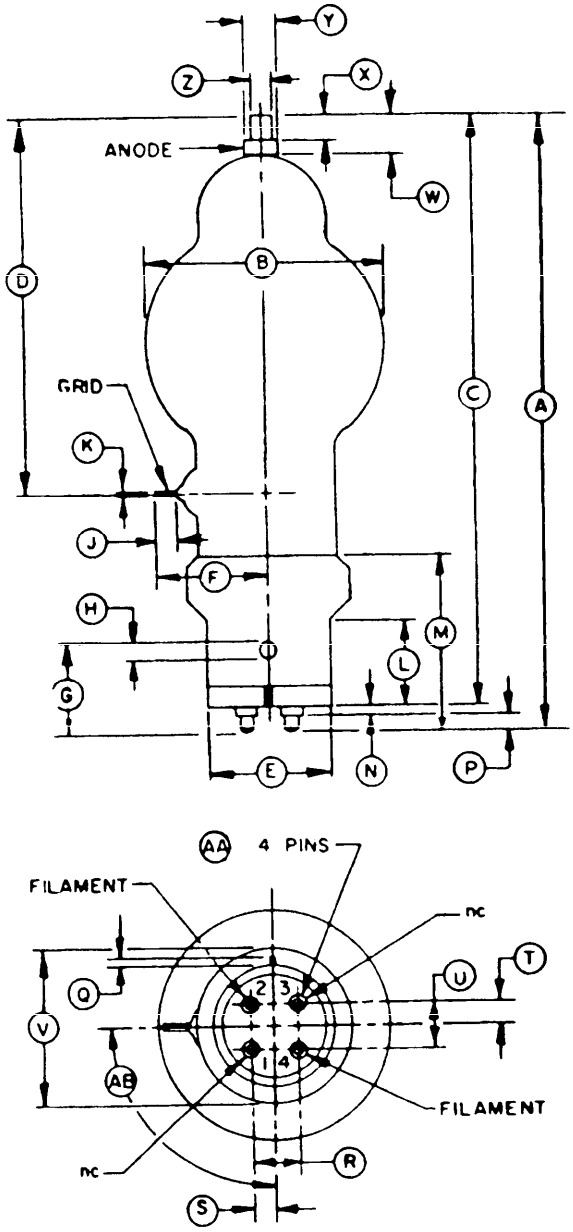
User activities:
Army - MU, WC
Navy - AS, OS, MC, CG, SH
Air Force - 11, 19

(D) Preparing activity: Army - EL

Agent: DSA - ES

(Project 5960-2842)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		NOTES
	Minimum	Maximum	
Quality conformance inspection, part 2			
A	9.625(244.48)	10.125(257.18)	
B		3.813 (96.85)	
D	5.688(144.48)	6.188(157.18)	
F	1.563 (39.70)	1.938 (49.23)	
J	.312 (7.92)		
Quality conformance inspection, part 3			3
C	9.375(238.13)	9.875(250.83)	
E		1.867 (47.42)	
H		.082 (2.08)	
K	.065 (1.65)	.071 (1.80)	
P	.250 (6.35)	.320 (8.13)	a
Q		.109 (2.77)	
V		2.500 (63.50)	
X	.328 (8.33)		
Z	.350 (8.89)	.365 (9.27)	
AA	.185 (4.70)	.189 (4.80)	
AB	80°	100°	
	Reference dimensions		
G	1.225(31.12)		a
L	1.938(49.23)		
M	2.719(69.06)		a
N	.031 (.79)		
R	.688(17.48)		
S	.344 (8.74)		
T	.344 (8.74)		
U	.688(17.48)		
W	.563(14.30)		
Y	.500(12.70)		

NOTE:
a. On finished tube, add 0.060 inch (1.52 mm) maximum for solder.

FIGURE 1. Outline drawing of electron tube type 250TH.

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-E-1/878D ELECTRON TUBE, POWER 1/ TYPE 250TH

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-PM 1068-68

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